Mehrab Hossain Shakib

+8801859640812 | shakibmehrab@gmail.com | mehrabhossain.portfolio | Mehrab Hossain Shakib | shakib-mehrab
Cumilla, Bangladesh

Objective

A CSE graduate from Comilla University and NASA Space Apps Challenge 2025 Global Finalist with hands on experience in full-stack, blockchain, and applied ML. Seeking an opportunity to contribute production ready solutions across web, decentralized, and AI-powered systems.

Technical Skills

- **Languages:** JavaScript, Python, C++, Solidity, Java, Rust
- **Web:** React, Next.js, Node.js, Express.js, Tailwind, REST API, GraphQL, Webhooks
- **App:** React Native, Expo
- **Database:** MySQL, PostgreSQL, MongoDB, Firebase, Supabase, Redis, ChromaDB
- **Blockchain/Web3:** Smart Contract, Hyperledger Fabric, Chainlink, Hardhat, IPFS, Solana, Web3.js, Ethers.js
- **Payment:** SSLCommerz, MetaMask
- **Core CS:** OOP, Data Structures, Algorithms, Problem Solving, Design Principle, SDL
- **ML:** Supervised Learning, Deep Learning, Ensemble Methods, Model Evaluation
- **AI/LLM:** LangChain, RAG, Hugging Face, Prompt Engineering
- **Tools & Platforms:** Linux, Git, Docker, CI/CD, Google Colab, Jupyter Notebook, Overleaf, n8n
- **Design:** Figma, Canva, Illustrator, Photoshop

Achievements

- Global Finalist — Nasa Space Apps Challenge 2025
- Finalist — AI Healthcare Hackathon 2026, Tashkent
- Finalist — SOLVIO AI Hackathon 2026
- Finalist — Hult Prize Comilla University 2025-26
- Finalist — Blockchain Olympiad Bangladesh 2024

Education

Comilla University *Feb 2022 – June 2026*
B.Sc. in Computer Science and Engineering *CGPA: 3.50 / 4.00*

Experience

Web Development Intern *Mar 2026 – Apr 2026*
Syntecxhub
- Delivered 2 production REST API modules within the first 3 weeks

Projects

- **HybridGuard — IoT Security Framework**
  Hyperledger Fabric, XGBoost, IPFS, LIME/SHAP, FL
  5-layer blockchain + federated learning system for IoT threat detection. Capstone research project under Dr. Mahmudul Hasan (SMIEEE).
- **Momentum — AI Productivity Tool**
  PERN, LangChain, ChromaDB, FastAPI, Docker
  AI-powered task automation platform with context aware scheduling.
- **Pathik — BD Travel Companion**
  React, Node.js, Firebase, PWA
  Offline-capable travel guide for Bangladesh with curated itineraries.
- **Tuition Track App**
  React Native, Expo, Firebase
  A React Native app for tutors and students to manage tuitions, classes, homework, and payments.
- **ChatDapp — Decentralized Chat**
  Next.js, Solidity, Hardhat, Ethers.js, MetaMask
  On-chain messaging DApp with wallet-based auth and no central server.
- **MintZillo — Real Estate DApp**
  React, Solidity, Hardhat, IPFS
  Tokenized real estate ownership and transfer on Ethereum (Sepolia tesetnet).
- **FundChain — Crowdfunding DApp**
  React, Solidity, Ethers.js, MetaMask, HardHat
  Contract enforced crowdfunding with on-chain fund release.
- **ASD Clinical Decision Support**
  PERN, FastAPI, Docker
  Human-in-the-loop VR therapy management system for ASD with AI-adaptive guidance, real-time session monitoring, and ABA-compliant reporting across therapist and guardian portals.
- **RoadGuardian AI**
  React, FastAPI, LangGraph, YOLOv8, Groq/Llama 3.1
  Offline first multi agent emergency coordination platform with RAG powered medical guidance, real time AI orchestration, and accessibility first UX for underserved regions.
- **NovaMind XR — AI Reality Composer**
  React, Three.js, Node.js, MongoDB, Gemini/Claude API
  WebXR platform that diagnoses science misconceptions via AI Socratic mentoring, real time 3D physics simulations, and a cognitive radar dashboard.
- **ZeroWaste — Sustainable Food Assistant**
  PERN, Prisma ORM, LangChain
  Food waste management app with intelligent inventory tracking and sustainability insights. Built with 3 developers at BUBT INNOVATEX Hackathon 2025.
- **Stellar Tales — Space Weather Education**
  MERN, REST API, NASA/NOAA Data
  Interactive children's story platform explaining space weather using real-time NASA and NOAA data. Built with 2 developers for NASA Space Apps Challenge 2025.

Certifications & Activities

Certifications & Participations
- Blockchain Foundation Course — Cyfrin Updraft
- Deep Learning — Coursera (2025)
- ICPC Regional Participant — 2023

Extracurricular Activities
- General Secretary — Comilla University IT Society
- General Secretary (Elected) — CSE Society, CSE, CoU
- Deputy Lead (Operations & Technical) — TEDx CoU
- Convener — BASIS Student Forum, CoU Chapter